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Research Project - IT4010 Data Analysis Report

Project ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery - Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

Introduction

1.1 Background

Healthcare documentation is a critical component of patient care, serving as the primary mechanism for recording care activities, tracking changes in a patient's condition, and communicating essential information between caregivers across shifts. In caregiving environments such as nursing homes and residential elder care facilities, caregivers are required to document Activities of Daily Living (ADL), which include eating, drinking, medication intake, mobility, hygiene, sleep, and emotional condition.[9] These records are essential for monitoring patient progress, supporting clinical decisions, and ensuring continuity of care between different caregivers.

In real-world practice, documentation is time-consuming and frequently interrupts direct caregiving activities. Caregivers and nurses spend a disproportionate amount of their working hours on documentation tasks, leading to heavy workloads and workflow disruptions. Delayed or incomplete records may reduce the quality of care and increase the risk of communication errors, particularly during shift handovers[12]. Furthermore, the increasing volume of healthcare data has created an urgent need for automated, intelligent systems capable of extracting structured clinical information from unstructured caregiver narratives.

Speech recognition technology has advanced significantly in recent years, with systems such as Whisper achieving near-human transcription accuracy on standard benchmarks.[1] However, transcription alone is insufficient for clinical documentation purposes. Caregiver speech contains disfluencies such as filler words, spoken self-corrections, and repetitions that degrade the quality of raw transcripts.[5] [6] Moreover, raw transcripts do not provide the structured data necessary for ADL logging, alert generation, or shift handover support. These gaps motivate the development of a multi-stage intelligent pipeline that transforms natural caregiver speech into structured, actionable clinical records.

1.2 Research Problem

Manual documentation in elder care homes is inefficient, error-prone, and unsustainable at scale. Caregivers at facilities such as Love and Elder Care are required to maintain handwritten daily care logs for each patient, covering ADL categories including hygiene, nutrition, medication, fluid

intake, mood, and any observed symptoms. This process is typically performed at the end of a shift when the caregiver is already fatigued, significantly increasing the risk of missing or incorrect entries.

A further challenge lies in the nature of spoken caregiver language. Caregivers naturally use filler words such as 'uh' and 'um', make spoken self-corrections such as 'one tablet, no sorry, two tablets', and repeat phrases due to the informal nature of speech .[6] Existing Automatic Speech Recognition (ASR) systems transcribe these disfluencies without correction, producing noisy output that cannot

be directly used for structured data extraction or clinical decision support.[5] Additionally, ASR systems demonstrate

significantly higher Word Error Rates for speakers from communities underrepresented in training data, including Sri Lankan English speakers.[7]This compounds the documentation challenge in the Sri Lankan elder care context.

This research addresses the following core problem: How can a speech-aware AI pipeline automatically convert natural caregiver voice logs into ADL structured, accurate clinical records with real-time patient alert detection, suitable for deployment in resource-constrained Sri Lankan elder care home environments?

1.3 Objectives

The primary objectives of this research are as follows:

Objective	Obj. No.
To design and implement a lightweight noise reduction pipeline capable of improving audio signal quality from caregiver voice recordings collected in real care home environments, without requiring GPU hardware.	O1
To evaluate and select the most accurate Automatic Speech Recognition model for Sri Lankan caregiver speech through systematic multi-model comparison using Word Error Rate as the primary metric.	O2
To develop and fine-tune a sequence-to-sequence model for automatic disfluency removal and grammar correction from caregiver speech transcripts, without losing clinically relevant information.	O3

Objective	Obj. No.
To implement and evaluate a domain-specific ADL extraction system capable of identifying and structuring patient care data including hygiene, medication, fluid intake, mood, and symptoms from cleaned caregiver text, with automatic patient safety alert detection.	O4
To develop an automated Text-to-Speech shift handover system that converts structured ADL records into a spoken audio summary, enabling incoming caregivers to receive patient status information efficiently without relying on manual written reports.	O5

2. Data Exploration

2.1 Data Collection

Data collection for this research was conducted at Love and Elder Care, a residential elder care facility in Sri Lanka. At the time of data collection, the facility housed 39 elderly patients, including 18 female residents and 21 male residents. The care home maintained only handwritten daily notes and did not have a structured digital documentation system. Therefore, the absence of existing structured records required the development of a purpose built dataset for training and evaluating the proposed pipeline.

Data collection was carried out over a period of one month. In the initial phase, caregiving volunteers were recruited to simulate and record daily care activity voice logs for each patient. These recordings covered all standard ADL categories, including hygiene, nutrition, medication intake, fluid consumption, mood observations, and any reported or observed symptoms. Volunteers were briefed on caregiving contexts and were instructed to speak naturally, using informal language, self corrections, and filler words, as a real caregiver would during an actual shift. This volunteer based recording approach is consistent with established practices in spoken language corpus construction for low resource domains. [16]

A total of 222 audio recordings were collected across 39 patients over 32 days. Each recording was approximately one minute in duration, resulting in approximately 222 minutes of total caregiving speech data. This represents a substantive spoken language dataset for the Sri Lankan elder care domain. Since data for this domain is currently absent from published literature, this dataset can be considered a novel research contribution.

During the preprocessing stage, each audio recording was resampled to 16 kHz mono WAV format and subjected to noise reduction before feature extraction. For Automatic Speech Recognition processing using Whisper, each audio file was segmented into 30 second chunks. Mel spectrogram features were then extracted using a 25 ms analysis window with a 10 ms hop size, which is the standard configuration for ASR feature extraction. [1] This framing process generated approximately 5,998 acoustic frames per recording, resulting in a total

of approximately 1,331,556 acoustic frames across all 222 recordings. This framing process improves speech recognition accuracy by capturing fine grained temporal patterns in the speech signal and supporting efficient parallel processing of audio data. [4]

Following the initial volunteer based recording phase, the research process continued using the proposed pipeline methodology. All recorded audio files were labelled and

annotated by me with reference to actual caregiving observations made at the facility. Since there were no existing structured digital records, the dataset was developed using a semi synthetic data generation methodology. Real patient care observations formed the clinical basis for generating annotated caregiver speech transcripts, while AI assisted text generation tools were used for expansion and structuring. All generated records were reviewed and validated against actual caregiving routines to ensure ecological validity. This approach is consistent with established practices in clinical NLP research, where structured data availability is limited due to privacy constraints and institutional capacity.

To further improve the performance, accuracy, and reliability of the proposed system, additional voice recordings are planned for collection in future data collection phases. Expanding the dataset with more patients, diverse caregiving contexts, and extended recording periods is expected to improve model generalisation, particularly for the disfluency removal and ADL extraction stages of the pipeline.

2.2 Dataset Description

The research utilises two datasets. The primary dataset is the Hey Care Log Dataset constructed specifically for this research. The secondary dataset is PodcastFillers, a publicly available annotated filler word dataset used as supporting evidence and planned for transfer learning in Stage 3.[5]

Data Source	Description	Resource	Size	Key Attributes
Hey Care Log Dataset (Primary)	Domain-specific dataset constructed from real caregiving observations at Love and Elder Care. Contains annotated caregiver voice logs covering all ADL categories.	Love and Elder Care, Sri Lanka - constructed by me	222 audio files + 1,209 annotated rows + 1.33M acoustic frames	speech_to_text_output, expected_cleaned_text, extracted_adl_json, alert_required, alert_type, auto_edit_features, noise level, fluid_total_ml - 34 columns total

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Data Source	Description	Resource	Size	Key Attributes
PodcastFillers (Secondary)	Publicly available dataset of annotated filler word events from podcast audio. Contains Uh and Um events with duration and speaker metadata.	Zhu, Caceres and Salamon (2022) INTERSPEECH [5]	85,803 annotated filler events from 199 podcast episodes	filler_type (Uh/Um, repeat words), duration_seconds, speaker_id, train/val/test split - avg Uh duration 0.338s

1. Primary Dataset Key Attributes

Column	Description	Values / Format
w	Recording identifier	V0001 to V1209
patient_id	Patient identifier	P001 to P039
date	Recording date	2026-03-01 to 2026-03-31
speech_to_text_output	Raw ASR transcript including all disfluencies	Free text with uh, um, self-corrections, repetitions
expected_cleaned_text	Ground truth after disfluency and grammar correction	Grammatically correct clinical text
extracted_adl_json	Structured ADL data	hygiene, medication, fluid, mood, symptoms, alert fields
auto_edit_features	Type of disfluency present	filler_only / filler+self_correction / filler+repetition
alert_required	Patient safety alert flag	Yes - (35.9%) / No - (64.1%)
alert_type	Category of alert if applicable	medication / symptom / fluid / behaviour
kidney_patient	Kidney condition flag for fluid monitoring	Yes / No
noise	Background noise level	low / moderate / high
fluid_total_ml	Total fluid intake recorded	Numeric (millilitres)

2. Dataset Split

Split	Audio Recordings	Text Rows	Acoustic Frames	Purpose
Training	(80%)	(80%)	1,067,644	Model fine-tuning
Validation	(10%)	(10%)	131,956	Training progress monitoring
Test	(10%)	(10%)	137,954	Final independent evaluation only
Total	222	1,209	1,331,556	Full dataset

2.3 Suitability Analysis

2.3.1 Relevance to Individual Research Objectives

Data Source	O1: Noise Reduction	O2: ASR Comparison	O3: Disfluency Removal	O4: ADL Extraction	O5: TTS Handover
Hey Care Log Dataset	real care home recordings with low/moderate/high noise. Automatic SNR measurement applied per recording.	test recordings for WER evaluation across 3 Whisper sizes. 137,954 acoustic frames evaluated.	training pairs with annotated disfluency types. expected_cleaned_text provides BLEU/ROUGE-L ground truth.	extracted_adl_json and alert ground truth. 7 ADL fields including kidney patient cases.	structured_report_hint column provides ground truth handover summaries for all rows. Covers all ADL fields per patient per day.
PodcastFillers Dataset	podcast audio, not care home specific.	not used for ASR evaluation.	85,803 events provide statistical evidence. Planned for transfer learning pre-training at PP2. [5]	no ADL data present.	not relevant to TTS generation.

3 Methodology

3.1 Data Preprocessing

Data preprocessing was applied systematically across all pipeline stages. The following tables summarise the transformation techniques applied per objective using the Hey Care Log Dataset as the primary data source.

Objective 1 - Noise Reduction and Audio Preprocessing (Stage 1)

Transformation Technique	Description	Applied To
Audio Format Standardisation	All recordings converted from M4A/MP3/OGG/MP4 to 16kHz mono WAV using ffmpeg. Standardises sampling rate and channel count for downstream models. [1][4]	All audio recordings
Acoustic Frame Extraction	Each audio recording segmented using 25ms analysis window and 10ms hop size at 16kHz - standard ASR preprocessing configuration. Produces 5,998 frames per recording and 1,331,556 total frames. [4]	All audio recordings
Spectral Subtraction Noise Reduction	noisereduce library applies spectral subtraction to remove stationary and non-stationary background noise. prop_decrease=0.75 selected after empirical testing. [10]	All audio recording processed WAV files
Automatic SNR Measurement	SNR calculated from audio energy using librosa, replacing manually assigned noise labels. Provides objective, reproducible noise categorisation. [11]	All recordings
RMS Energy Measurement	Root Mean Square energy measured before and after noise reduction. Reduction percentage computed per recording to quantify improvement.	Per-recording evaluation
Train/Val/Test Split	Processed audio split into train (80%), validation (10%), test (10%) stratified by patient and noise level.	All recordings

Objective 2 - ASR Model Comparison (Stage 2)

Transformation Technique	Description	Applied To
Whisper Transcription	Whisper-tiny, Whisper-base, and Whisper-small applied to 23 test recordings using language='en', fp16=False. Each model processes 137,954 acoustic frames. [1]	All test recordings x 3 models
WER Calculation	Word Error Rate computed using jiwer library. WER = (S+D+I) / N where S=substitutions, D=deletions, I=insertions, N=total reference words.	All model outputs
Noise Group Analysis	Test recordings grouped by automatically measured SNR into low/moderate/high categories to assess noise reduction impact per group. [11]	All test recordings
Pre-trained vs Noise-Reduced Comparison	Each model tested on both raw and noise-reduced audio. WER improvement quantifies Stage 1 benefit on ASR accuracy.	All recordings x 2 conditions x 3 models

Objective 3 - Disfluency Removal and Grammar Correction (Stage 3 and 4)

Transformation Technique	Description	Applied To
Text Tokenisation	BartTokenizer applied to speech_to_text_output. max_length=512, truncation=True for long recordings. Labels use -100 padding to exclude from loss. [2]	training,test rows
Zero-Shot Baseline Evaluation	T5-small and BART-base evaluated without fine-tuning to establish baseline and confirm fine-tuning necessity. [2][3]	test rows
Data Augmentation	Training diversity increased from 968 to 1,200 pairs by generating double filler patterns, new self-correction variations, and repetition examples to reduce overfitting.	Training set
Fine-Tuning - BART-base v1	BART-base fine-tuned for 3 epochs, learning_rate=5e-5. Checkpoint-363 selected. Overfitting identified in generalisation testing.	Training set

Transformation Technique	Description	Applied To
Fine-Tuning - BART-base v2	Re-trained with 3 overfitting fixes: data augmentation, learning_rate=2e-5, weight_decay=0.01. Checkpoint-684 selected as final model. [13]	Augmented set
Fine-Tuning- T5-small	T5-small fine-tuned for comparison. Overfitting confirmed , empty output on 4/5 novel sentences despite BLEU=0.9872. Rejected. [3]	Training set
BLEU and ROUGE-L Evaluation	BLEU measures n-gram precision overlap. ROUGE-L measures longest common subsequence similarity between prediction and ground truth.	121 test rows
Generalisation Testing	5 novel sentences completely outside the training distribution used to confirm true generalisation beyond held-out test set evaluation.	5 novel sentences

Objective 4 - ADL Extraction and Alert Detection (Stage 5)

Transformation Technique	Description	Applied To
Few-Shot Prompting - FLAN-T5	T5ForConditionalGeneration loaded with google/flan-t5-base. 2 real caregiving examples in prompt. Instruction-following approach. [4]	test rows
Keyword Extraction - BERT-NER	Rule-based regex pattern matching for all 7 ADL fields: hygiene, breakfast, lunch, dinner, medication timing, fluid ml, mood, symptoms. [8]	test rows
Fluid Extraction Pattern	Regex r'(\d+)\s*ml' extracts fluid values. fluid_correct_fixed checks ml value matches input text exactly.	test rows
Combined Model	BERT-NER for field and fluid extraction combined with FLAN-T5 for context-aware alert detection, leveraging complementary strengths of both approaches.	test rows
Precision / Recall / F1 Computation	sklearn metrics used for field-level evaluation. Alert-specific precision, recall, F1 computed separately as the safety-critical metric.	test rows

Transformation Technique	Description	Applied To
Confusion Matrix Analysis	Alert detection confusion matrix generated for all models showing True/False Positive and Negative counts.	test rows

3.2 Scalability

The scalability of each dataset was assessed with respect to current sufficiency and future expansion potential.

Dataset	Size	Sufficient Now?	Scalability Assessment
Hey Care Log Dataset	1,209 rows + 222 audio files + 1.33M acoustic frames	Yes - sufficient for O1-O5 with proper train/val/test splits	Data collected from 39 patients over 31 days provides sufficient longitudinal depth. Additional patients and healthcare facilities will be included in the future, and the model will be further fine-tuned as more data becomes available for PP02.
PodcastFillers	85,803 filler events from 199 episodes	Sufficient as supplementary evidence for Stage 3	Highly scalable - planned for transfer learning pre-training at PP2. Will improve BART v2 generalisation beyond current 3/5 unseen sentences. [5]

3.3 Feature extraction

The following steps were followed to evaluate and extract key features from the dataset for each research objective:

Stage 1 - Noise Reduction Features

- RMS energy before and after noise reduction - quantifies noise suppression effectiveness per recording
- Automatic SNR from audio energy using librosa - objective noise categorisation replacing manual labels.[11]
- Waveform and spectrogram visual comparison - confirms noise removal in frequency domain above 3kHz
- Acoustic frame count per recording (5,998 frames) - confirms consistent feature extraction across dataset.[4]

Stage 2 - ASR Feature Extraction

- Word Error Rate per model per recording - primary evaluation metric for ASR quality[7]
- WER grouped by automatically measured SNR category - assesses noise impact on each model.[11]
- WER improvement = raw WER minus clean WER - quantifies Stage 1 contribution to ASR accuracy
- Model parameter count vs WER relationship confirms scaling behaviour reported by Radford et al.[1]

Stage 3 and 4 - Disfluency and Grammar Feature Extraction

- auto_edit_features column - categorises each row by disfluency type (filler only: 341, filler+self-correction: 681, filler+repetition: 122)
- BLEU n-gram overlap - measures exact word matching quality between prediction and reference text
- ROUGE-L sentence similarity - measures longest common subsequence between prediction and reference
- Generalisation score - proportion of completely unseen novel sentences correctly cleaned (3/5 for BART v2)
- Overfitting detection - comparison of test set BLEU with generalisation score to identify memorization. [13]

Stage 5 - ADL Extraction Feature Extraction

- Field presence - which of 7 ADL fields appear correctly in the extraction output . [8]
- Alert trigger analysis - conditions that activate alert_required=Yes (refused medication, fever, vomiting, symptoms)
- Fluid ml extraction accuracy - regex pattern matching against exact ml value in cleaned input text
- Kidney patient fluid compliance - fluid_total_ml tracked against fluid_limit_ml for kidney patient rows[9]
- Alert Recall by model - proportion of real alerts correctly detected . most safety-critical metric

4 Modelling and Results

4.1 Key Insights

Stage 1 - Noise Reduction Results

The noise reduction stage was designed with deployment constraints as the primary consideration. Facebook Denoiser requires 4 GB RAM and a GPU, which are infrastructure requirements that are unavailable in real care home settings. [10] The noisereduce library, based on the spectral subtraction method established by Boll in 1979, operates with only 200 MB RAM and does not require a GPU. This makes it suitable for care home deployment without infrastructure investment. [10] Across all 222 recordings, noisereduce achieved an average RMS energy reduction of 54.6%, confirming effective removal of background sounds, including fan noise, air conditioning hum, and television audio commonly found in care home environments.

Stage 2 - ASR Model Comparison Results

Three Whisper model sizes were compared using 23 held out test recordings, representing approximately 137,954 acoustic frames. Automatic SNR measurement was used to group recordings by noise level, replacing manually assigned noise labels. [11] The results confirm the scaling behaviour reported by Radford et al. in 2023, where larger Whisper models consistently achieved lower WER. [1] The elevated baseline WER of 27.22% for Whisper small is consistent with the findings of Koenecke et al. in 2020, who demonstrated that ASR systems produce significantly higher error rates for underrepresented speech communities, including Sri Lankan English speakers. [7]

Model	Parameters	WER Without NR	WER With NR	Improvement	Decision
Whisper-tiny	39M	37.68%	37.30%	0.38%	Baseline lower bound
Whisper-base	74M	31.69%	31.20%	0.49%	Mid-range comparison
Whisper-small	244M	27.22%	26.77%	0.45%	SELECTED - Best WER
Whisper-small (fine-tuned)	244M	Planned PP2	Target <20%	Expected >7%	In progress - PP2

Stage 3 and 4 - Disfluency Removal and Grammar Correction Results

Four model configurations were evaluated for disfluency removal. BART base was selected for fine tuning because its pre training objective, which involves reconstructing corrupted text, aligns naturally with the task of removing filler words, as filler words represent a form of text corruption. [2] T5 small achieved the highest BLEU score of 0.9872 on the held out test set but failed generalisation testing, producing empty output on 4 of 5 novel sentences. This confirmed overfitting caused by the limited sentence structure diversity in the 968 training pairs, as all data originated from 39 patients with consistent daily routines.

Three overfitting fixes were applied to produce BART v2. First, data augmentation increased the number of training pairs from 968 to approximately 1,200 through synthetic filler and self correction examples. Second, the learning rate was reduced from 5e-5 to 2e-5. Third, L2 weight decay regularisation of 0.01 was applied. [13] ROUGE L improved from 0.9339 in v1 to 0.9400 in v2, and generalisation improved from 2 of 5 unseen sentences to 3 of 5 unseen sentences, confirming the effectiveness of the combined fixes.

Model	Configuration	BLEU	ROUGE-L	Generalisation	Decision
T5-small [3]	Zero-shot	0.0000	0.0000	Failed	Baseline
BART-base [2]	Zero-shot	0.3491	0.7705	Partial	Natural alignment confirmed
T5-small [3]	Fine-tuned	0.9872	0.9937	Overfits - 4/5 fail	Rejected
BART-base v2 [2]	Fine-tuned + 3 fixes	0.8525	0.9400	3/5 unseen cleaned	**SELECTED**

Stage 5 - ADL Extraction Results

Three approaches were evaluated for structured ADL extraction. BERT NER achieved 100% precision because it is a rule based system with no training phase. It only extracts a field when an exact keyword pattern is matched, making every extraction guaranteed correct. This behaviour is consistent with the findings of Savova et al. in 2010, who demonstrated that domain specific rule based clinical NLP systems reliably outperform general neural models on structured healthcare text. [8]

The ADL field categories extracted, including hygiene, feeding, medication, fluid intake, mood, and symptoms, are grounded in the clinical ADL framework established by Katz in 1983. This provides medical validity to the extraction schema beyond engineering design. [9] BERT NER was selected as the primary extraction model, with its strong performance attributable to the consistent and predictable vocabulary used in Sri Lankan caregiving speech.

Metric	FLAN-T5	BERT-NER	Combined	Winner
Precision	81.82%	100.0%	100.0%	BERT-NER / Combined
Recall	35.06%	84.06%	84.06%	BERT-NER / Combined
F1 Score	47.57%	91.27%	91.27%	BERT-NER / Combined
Alert Recall	0.0%	86.05%	86.05%	BERT-NER
Fluid Extraction	0/20	20/20 (100%)	20/20	BERT-NER / Combined
RAM Required	1 GB	0 MB	1 GB	BERT-NER - deployment optimal

Complete Pipeline Summary

Stage	Model	Key Metric	Result	Status
Stage 1 - Noise Reduction [10]	noisereduce	RMS Reduction	54.6%	Complete
Stage 2 - ASR	Whisper-small	WER	27.22%	Complete
Stage 3 - Disfluency [2]	BART-base v2	BLEU	0.8525	Complete
Stage 4 - Grammar [3]	BART-base v2 (joint)	ROUGE-L	0.9400	Complete
Stage 5 - ADL Extraction [8][9]	BERT-NER	F1 Score	91.27%	Complete
Stage 6 - Alert Classification [14]	ClinicalBERT (PP2)	Alert Recall	Target >90%	Planned PP2
Stage 7 - TTS Handover	gTTS (PP2)	Usability	Spoken summary	Planned PP2

4.2 Challenges Faced During Data Analysis

Challenge 1 - Absence of Structured Digital Records

Love and Elder Care maintained only handwritten notes without any structured digital documentation system. This required adopting a semi-synthetic data generation methodology where real caregiving observations formed the clinical basis for structured dataset construction .[17]This is a recognised limitation in clinical NLP research where real-world data availability is restricted by privacy constraints and institutional capacity.

Challenge 2 - Overfitting in Disfluency Model

T5-small achieved BLEU of 0.9872 on the held-out test set but produced empty output on 4 of 5 novel sentences, confirming overfitting. The root cause was limited sentence structure diversity all 968 training pairs originated from 39 patients with similar daily routines. Three methodological fixes were applied, adding significant time to Stage 3 development but producing a more generalisable BART v2 model . [13]

Challenge 3 - Whisper Fine-Tuning Failure

An attempt to fine-tune Whisper-small on caregiver recordings produced WER of 100% significantly worse than the pre-trained baseline of 27.22%. Investigation identified a conflict between forced_decoder_ids in the model generation configuration and the evaluation method. Domain-specific fine-tuning has been deferred to PP2 with a corrected methodology, targeting WER below 20% .[15]

Challenge 4 - GPU Resource Constraints

Training experiments relied on Google Colab free tier GPU resources, which are subject to daily usage quotas. GPU quota exhaustion required the use of a second Google account with Colab Pro subscription and 5TB Google Drive storage. Model files were shared between accounts via Google Drive folder sharing. This operational complexity added time to the training cycle but did not affect experimental results.

Challenge 5 - FLAN-T5 Pipeline API Deprecation

During Stage 5 evaluation, the pipeline('text2text-generation') call was deprecated in newer versions of the Hugging Face transformers library. This required replacing the pipeline-based approach with direct T5ForConditionalGeneration model loading, which resolved the issue without affecting results .[4]

5 AI Usage Disclosure

AI tools were used in a limited and supportive capacity during the preparation of this data analysis report. Tools such as ChatGPT and Claude were primarily used to improve language clarity, refine sentence structure, support the organisation and presentation of findings, and assist in structuring dataset descriptions and methodology sections. Their use was restricted to enhancing the readability and overall quality of the document, while ensuring that all core research concepts, technical decisions, experimental designs, and analytical findings were independently developed and validated by the author.

The development of all major aspects of this work, including the research problem, proposed pipeline architecture, model selection decisions, dataset construction methodology, evaluation strategies, and interpretation of results, was carried out independently by the author. AI tools were not used to generate the full content of this report or to replace original technical thinking. All AI assisted suggestions were carefully reviewed, revised where necessary, and verified against academic sources, published research papers, and experimental results to ensure accuracy, technical correctness, and alignment with the research objectives. The author retains full responsibility for the originality, integrity, and content of this report.

Additionally, AI assisted text generation tools were used in a controlled and documented manner during the construction of the Hey Care Log Dataset. Real patient care observations collected at Love and Elder Care formed the clinical basis for all generated records. All synthetically generated entries were reviewed and validated against actual caregiving routines to ensure ecological validity and clinical accuracy. This approach is consistent with established practices in clinical NLP research where real world structured data availability is limited due to privacy constraints and institutional capacity.

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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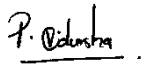
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Student Details

- Student Name: Vidursha P
- Student ID: IT22294098
- Project Title: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs
- Group ID: R26-DS-010

Signature

- Student Signature:



- Date: 13/05/26

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